DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-300-02 — Pull, clean and test the tube bundle of E-103

Area	A300 Exchangers and drum	**Task**	XB Exchanger bundle pull, clean and tube-side test
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	**E-103** — E-103 shell-and-tube exchanger 36" ID × 192" — E 87'-0" N 25'-0" EL 106'-6"		
Equipment	E-103	**Lines**	4"-P150-CWS-3005, 4"-P150-CWR-3006, 6"-P300-P-1011, 6"-P300-P-1012, 1"-P150-V-6027, 1"-P150-D-6028, 3/4"-P300-I-6029
Isometrics	ISO-4"-P150-CWS-3005, ISO-4"-P150-CWR-3006, ISO-6"-P300-P-1011, ISO-6"-P300-P-1012, ISO-1"-P150-V-6027, ISO-1"-P150-D-6028, ISO-3/4"-P300-I-6029 (generated); owner's drawings ISO-CWS-3005, ISO-CWR-3006, ISO-P-1011, ISO-P-1012, ISO-V-6027, ISO-D-6028, ISO-I-6029		
Permits	General work permit, Confined-space entry permit, Lifting plan		

Reason

Fouling: tube bundle to be pulled, cleaned and tested.

Scope of work

1. Isolate E-103 per the isolation list (tube and shell sides); drain and vent.
2. Remove the channel cover; pull the tube bundle to the cleaning bay.
3. Clean the tubes and shell side; inspect visually; thickness at the TMLs.
4. Reinstall the bundle with new gaskets; tube-side hydrostatic test at 210 psi (SW-TEST-1).
5. Remove blinds; restore the line-up.

Affected items: E-103

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-CWS-3005	4"-P150-CWS-3005	4"-P150-CWS-3005	1	0 error(s), 1 warning(s) — see below
ISO-CWR-3006	4"-P150-CWR-3006	4"-P150-CWR-3006	2	0 error(s), 2 warning(s) — see below
ISO-P-1011	6"-P300-P-1011	6"-P300-P-1011	2	0 error(s), 3 warning(s) — see below
ISO-P-1012	6"-P300-P-1012	6"-P300-P-1012	2	0 error(s), 2 warning(s) — see below
ISO-V-6027	1"-P150-V-6027	1"-P150-V-6027	1	agrees with the model
ISO-D-6028	1"-P150-D-6028	1"-P150-D-6028	1	agrees with the model
ISO-I-6029	3/4"-P300-I-6029	3/4"-P300-I-6029	1	agrees with the model

ISO-CWS-3005 (PDF) · ISO-CWR-3006 (PDF) · ISO-P-1011 (PDF) · ISO-P-1012 (PDF) · ISO-V-6027 (PDF) · ISO-D-6028 (PDF) · ISO-I-6029 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-CWS-3005	1	warning	weld 4"-P150-CWS-3005/W1 (E 86'-0" N 0" EL 123'-6 11/16", sheet 1) has no label
ISO-CWR-3006	2	warning	weld 4"-P150-CWR-3006/W9 (E 84'-5 5/16" N 25'-0" EL 126'-6", sheet 2) has no label
ISO-CWR-3006	2	warning	weld 4"-P150-CWR-3006/W12 (E 85'-0" N 1'-6" EL 125'-11 5/16", sheet 2) has no label
ISO-P-1011	2	warning	W14 (6"-P300-P-1011/W14, sheet 2) is labelled again on sheet 1; it is 162.0 in from the nearest break
ISO-P-1011	2	warning	weld 6"-P300-P-1011/W20 (E 89'-0" N 25'-0" EL 110'-2 1/2", sheet 2) has no label
ISO-P-1011	1	warning	branch weld 3/4"-P300-I-7011/W1 (E 68'-11" N 6" EL 120'-0", sheet 1) has no label
ISO-P-1012	2	warning	W10 (6"-P300-P-1012/W10, sheet 2) is labelled again on sheet 1; it is 70.5 in from the nearest break
ISO-P-1012	1	warning	branch weld 3/4"-P300-I-7012/W1 (E 99'-6" N 21'-0" EL 113'-11", sheet 1) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
E-103	equipment	E-103 shell-and-tube exchanger 36" ID × 192"		E 87'-0" N 25'-0" EL 106'-6"

Tag	Kind	Description	Line	Location
4"-P150-CWS-3005/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWS-3005	E 86'-0" N 25'-0" EL 109'-3-13/16"
E-103-N1	nozzle	Nozzle E-103-N1 (tube in, 4" P150) on E-103		E 86'-0" N 25'-0" EL 108'-8-1/16"
4"-P150-CWR-3006/A2	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWR-3006	E 86'-0" N 25'-0" EL 103'-8-3/16"
E-103-N2	nozzle	Nozzle E-103-N2 (tube out, 4" P150) on E-103		E 86'-0" N 25'-0" EL 104'-3-15/16"
6"-P300-P-1011/B2	item	VALVE, GATE, WEDGE, OS&Y 6" P300	6"-P300-P-1011	E 89'-0" N 25'-0" EL 109'-3-7/16"
E-103-N3	nozzle	Nozzle E-103-N3 (shell in, 6" P300) on E-103		E 89'-0" N 25'-0" EL 108'-8-5/16"
6"-P300-P-1012/A2	item	VALVE, GATE, WEDGE, OS&Y 6" P300	6"-P300-P-1012	E 101'-0" N 25'-0" EL 103'-8-9/16"
E-103-N4	nozzle	Nozzle E-103-N4 (shell out, 6" P300) on E-103		E 101'-0" N 25'-0" EL 104'-3-3/4"
1"-P150-V-6027/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-V-6027	E 100'-0" N 25'-0" EL 109'-1/2"
E-103-N5	nozzle	Nozzle E-103-N5 (vent, 1" P150) on E-103		E 100'-0" N 25'-0" EL 108'-5-11/16"
1"-P150-D-6028/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-D-6028	E 89'-0" N 25'-0" EL 103'-11-1/2"
E-103-N6	nozzle	Nozzle E-103-N6 (drain, 1" P150) on E-103		E 89'-0" N 25'-0" EL 104'-6-3/8"
3/4"-P300-I-6029/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-I-6029	E 95'-0" N 27'-6-1/4" EL 106'-6"
E-103-N7	nozzle	Nozzle E-103-N7 (instrument, 0.75" P300) on E-103		E 95'-0" N 26'-11-9/16" EL 106'-6"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P150-CWS-3005/B2	4"-P150-CWS-3005	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3005/B2) on E-103-N1	E 86'-0" N 25'-0" EL 109'-3-13/16"		
2	blind	E-103-N1	4"-P150-CWS-3005	Fit a blind between 4"-P150-CWS-3005/B2 and the nozzle flange E-103-N1	E 86'-0" N 25'-0" EL 108'-8-1/16"		
3	valve	4"-P150-CWR-3006/A2	4"-P150-CWR-3006	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWR-3006/A2) on E-103-N2	E 86'-0" N 25'-0" EL 103'-8-3/16"		
4	blind	E-103-N2	4"-P150-CWR-3006	Fit a blind between 4"-P150-CWR-3006/A2 and the nozzle flange E-103-N2	E 86'-0" N 25'-0" EL 104'-3-15/16"		
5	valve	6"-P300-P-1011/B2	6"-P300-P-1011	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1011/B2) on E-103-N3	E 89'-0" N 25'-0" EL 109'-3-7/16"		
6	blind	E-103-N3	6"-P300-P-1011	Fit a blind between 6"-P300-P-1011/B2 and the nozzle flange E-103-N3	E 89'-0" N 25'-0" EL 108'-8-5/16"		
7	valve	6"-P300-P-1012/A2	6"-P300-P-1012	Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1012/A2) on E-103-N4	E 101'-0" N 25'-0" EL 103'-8-9/16"		
8	blind	E-103-N4	6"-P300-P-1012	Fit a blind between 6"-P300-P-1012/A2 and the nozzle flange E-103-N4	E 101'-0" N 25'-0" EL 104'-3-3/4"		
9	valve	1"-P150-V-6027/2	1"-P150-V-6027	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6027/2) on E-103-N5	E 100'-0" N 25'-0" EL 109'-1/2"		
10	blind	E-103-N5	1"-P150-V-6027	Fit a blind between 1"-P150-V-6027/2 and the nozzle flange E-103-N5	E 100'-0" N 25'-0" EL 108'-5-11/16"		
11	valve	1"-P150-D-6028/2	1"-P150-D-6028	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6028/2) on E-103-N6	E 89'-0" N 25'-0" EL 103'-11-1/2"		
12	blind	E-103-N6	1"-P150-D-6028	Fit a blind between 1"-P150-D-6028/2 and the nozzle flange E-103-N6	E 89'-0" N 25'-0" EL 104'-6-3/8"		
13	valve	3/4"-P300-I-6029/2	3/4"-P300-I-6029	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6029/2) on E-103-N7	E 95'-0" N 27'-6-1/4" EL 106'-6"		
14	blind	E-103-N7	3/4"-P300-I-6029	Fit a blind between 3/4"-P300-I-6029/2 and the nozzle flange E-103-N7	E 95'-0" N 26'-11-9/16" EL 106'-6"		

Inspections

Id	Inspection	Criteria	Basis
WO-300-02-I1	Internal visual examination of the shell, channel and tube bundle	Findings recorded against the owner's integrity criteria	SW-NDE-4
WO-300-02-I2	Ultrasonic thickness at the 12 TMLs listed	Each reading above the owner's minimum wall	SW-WO-6

Thickness monitoring locations (SW-WO-6)

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-300-02-TML01	E-103	Nozzle neck E-103-N1	E 86'-0" N 25'-0" EL 108'-8-1/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-02-TML02	E-103	Nozzle neck E-103-N2	E 86'-0" N 25'-0" EL 104'-3-15/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-02-TML03	E-103	Nozzle neck E-103-N3	E 89'-0" N 25'-0" EL 108'-8-5/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-02-TML04	E-103	Nozzle neck E-103-N4	E 101'-0" N 25'-0" EL 104'-3-3/4"		top: ____	side: ____	bottom: ____	side: ____
WO-300-02-TML05	E-103	Nozzle neck E-103-N5	E 100'-0" N 25'-0" EL 108'-5-11/16"		top: ____	side: ____	bottom: ____	side: ____

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-300-02-TML06	E-103	Nozzle neck E-103-N6	E 89'-0" N 25'-0" EL 104'-6-3/8"		top: ____	side: ____	bottom: ____	side: ____
WO-300-02-TML07	E-103	Nozzle neck E-103-N7	E 95'-0" N 26'-11-9/16" EL 106'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-300-02-TML08	E-103	Left head centre	E 86'-2-3/4" N 25'-0" EL 106'-6"		centre: ____			
WO-300-02-TML09	E-103	Right head centre	E 103'-9-1/4" N 25'-0" EL 106'-6"		centre: ____			
WO-300-02-TML10	E-103	Shell at 0.000" from the left tangent	E 87'-0" N 25'-0" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-300-02-TML11	E-103	Shell at 6'-0" from the left tangent	E 93'-0" N 25'-0" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-300-02-TML12	E-103	Shell at 12'-0" from the left tangent	E 99'-0" N 25'-0" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____

Test — Hydrostatic test (tube side)

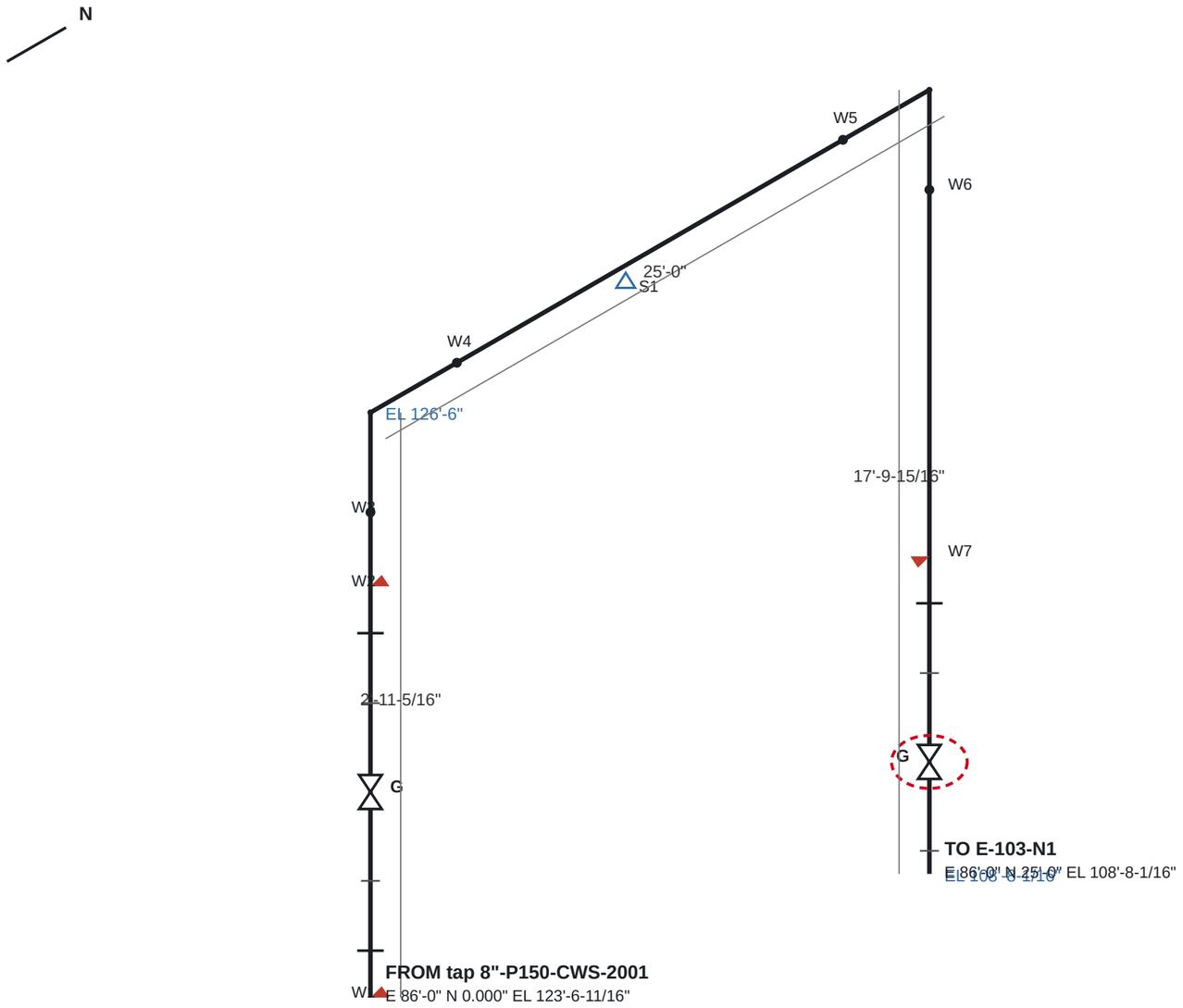
Description	Tube side of the reinstalled bundle
Pressure	210 psi
Hold	30 minutes, no leak, no pressure drop
Basis	SW-TEST-1

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Equipment vented, purged and gas-tested for entry; confined-space entry permit issued	Owner entry procedure	Atmosphere safe for entry per owner limits	Gas test record, entry permit	H	H	—				
3	Channel cover removed; bundle pulled and cleaned	Owner procedure	Surfaces clean enough to inspect	Daily report	S	S	—				
4	Internal visual examination	SW-NDE-4	Findings recorded	Inspection report with photographs	W	H	W				
5	Ultrasonic thickness at 12 TMLs	SW-WO-6	Above the owner's minimum wall	Thickness record per TML	S	R	R				
6	Owner's decision on findings (repair, monitor or accept)	Owner integrity procedure	Decision recorded	Signed disposition	R	H	R				
7	Closed with new gaskets on every opened joint; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
8	Tube-side hydrostatic test at 210 psi, 30 minutes	SW-TEST-1	No leak, no pressure drop	Test record	H	H	H				
9	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
10	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P150-CWS-3005 — Pull, clean and test the tube bundle of E-103



BILL OF MATERIALS

1	FLANGE WELD-NECK 4" P150 RAISED FACE	3
2	GASKET 4" P150 COMPRESSED SHEET	4
3	VALVE, GATE, WEDGE, OS&Y 4" P150	2
4	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	4
5	PIPE 4" P150 BEVELLED ENDS	39'-7-5/16"
6	ELBOW 90° R = 5/3 OD 4" P150	2

WELDS 7 (3 FIELD) · SUPPORTS 1 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7F

4"-P150-CWS-3005

CLASS W150 — Water (cooling and process)
SERVICE CWS · 4" · P150 · DESIGN 150 psi
tap 8"-P150-CWS-2001 → E-103-N1
PIPE 39'-7-5/16" · WEIGHT 693 lb

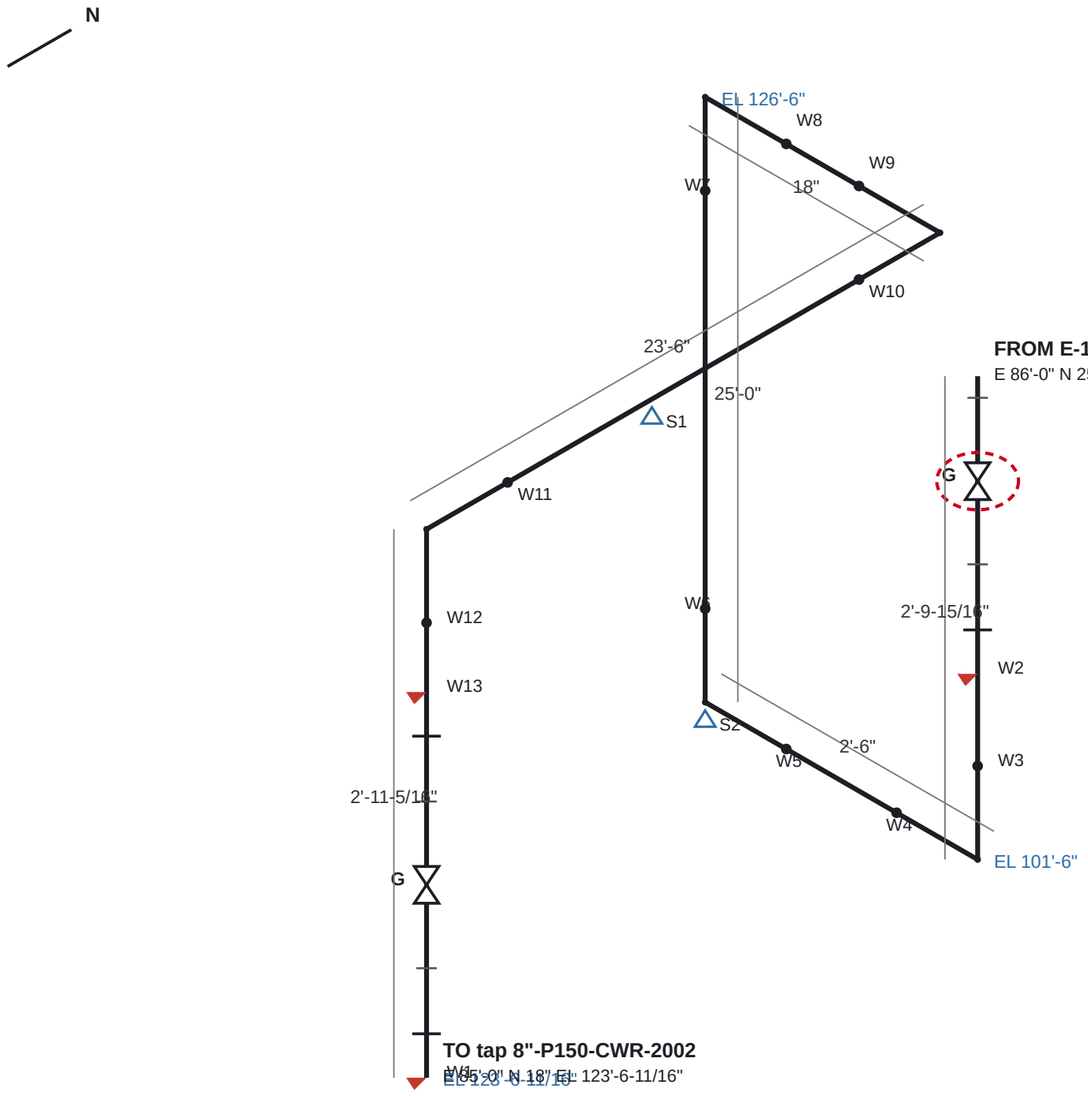
DWG ISO-4"-P150-CWS-3005 REV 0 WO-300-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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Isometric ISO-4"-P150-CWR-3006

ISOMETRIC — 4"-P150-CWR-3006 — Pull, clean and test the tube bundle of E-103



BILL OF MATERIALS

1	GASKET 4" P150 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 4" P150	2
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	4
4	FLANGE WELD-NECK 4" P150 RAISED FACE	3
5	PIPE 4" P150 BEVELLED ENDS	48'-9-5/16"
6	ELBOW 90° R = 5/3 OD 4" P150	5

WELDS 13 (3 FIELD) · SUPPORTS 2 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F

4"-P150-CWR-3006

CLASS W150 — Water (cooling and process)
SERVICE CWR · 4" · P150 · DESIGN 150 psi
E-103-N2 → tap 8"-P150-CWR-2002
PIPE 48'-9-5/16" · WEIGHT 783 lb

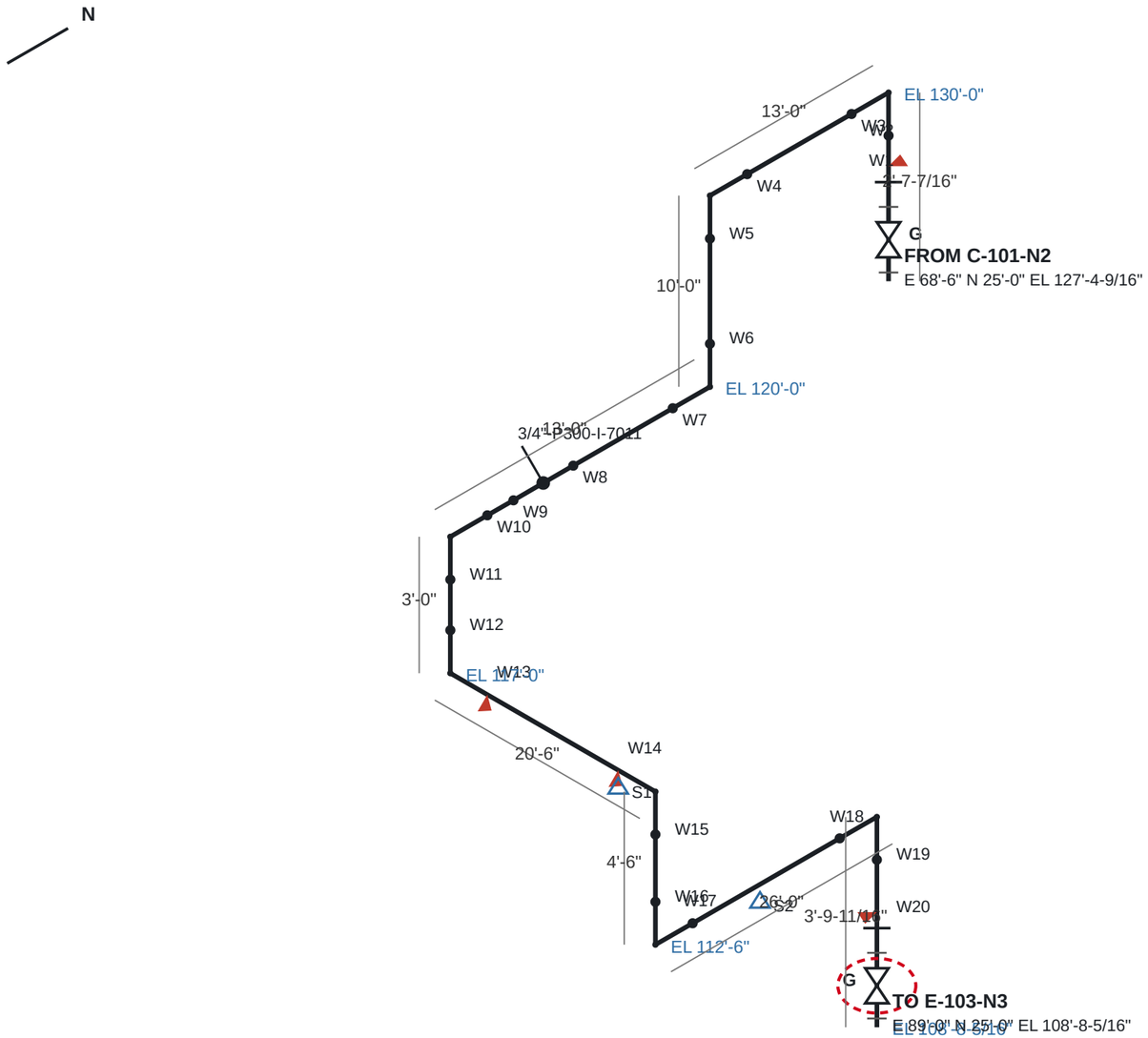
DWG ISO-4"-P150-CWR-3006 REV 0 WO-300-02

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Isometric ISO-6"-P300-P-1011

ISOMETRIC — 6"-P300-P-1011 — Pull, clean and test the tube bundle of E-103



BILL OF MATERIALS

1	GASKET 6" P300 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P300	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300	4
4	FLANGE WELD-NECK 6" P300 RAISED FACE	2
5	PIPE 6" P300 BEVELLED ENDS	79'-2-13/16"
6	ELBOW 90° R = 5/3 OD 6" P300	8
7	TEE REDUCING 6" × 3/4" P300	1

WELDS 20 (4 FIELD) · SUPPORTS 2 · SPOOLS 5

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F W14F W15 W16 W17 W18 W19 W20F

6"-P300-P-1011

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
C-101-N2 → E-103-N3
PIPE 79'-2-13/16" · WEIGHT 1677 lb

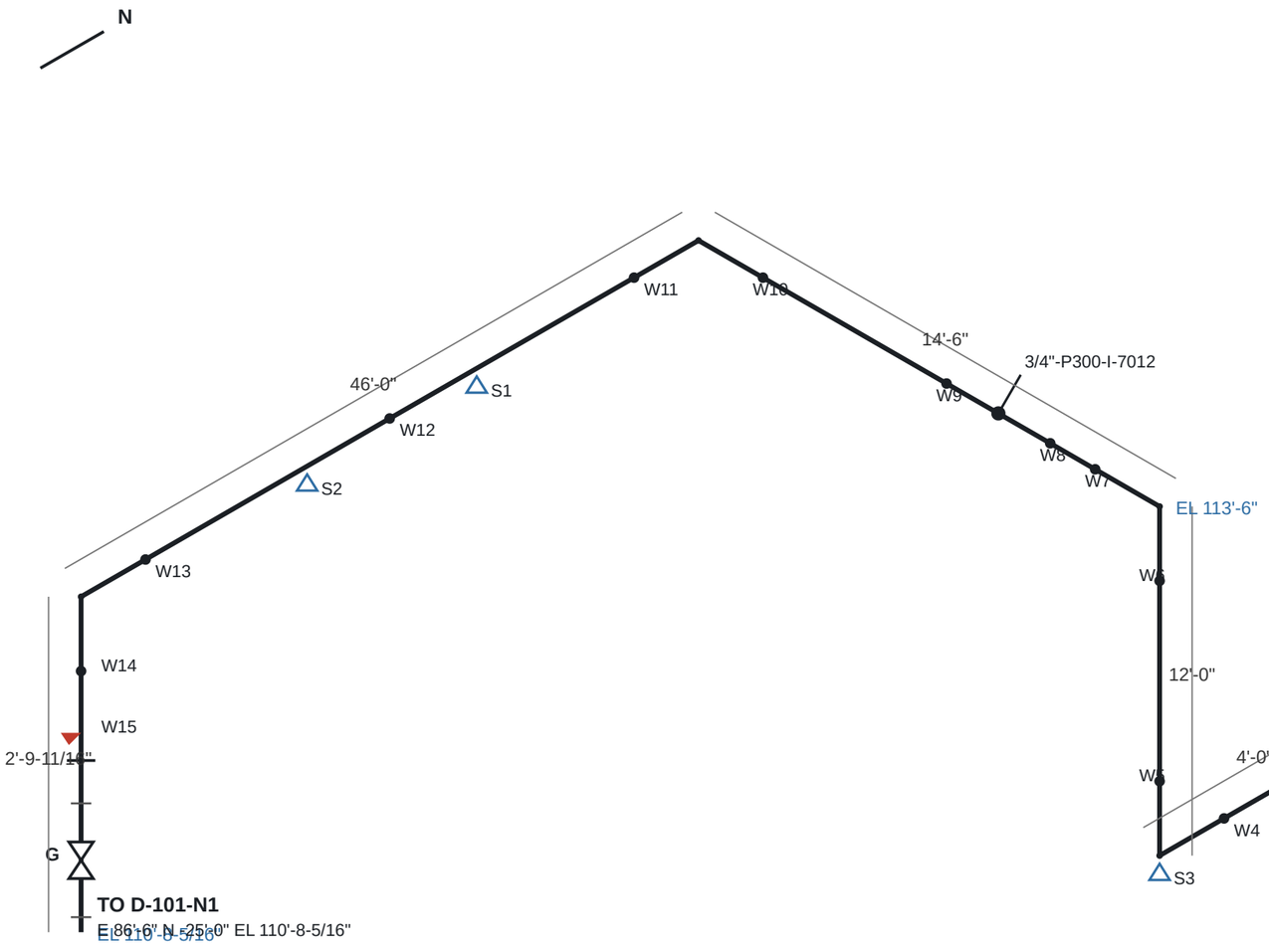
DWG ISO-6"-P300-P-1011 REV 0 WO-300-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-6"-P300-P-1012

ISOMETRIC — 6"-P300-P-1012 — Pull, clean and test the tube bundle of E-103



BILL OF MATERIALS

1	GASKET 6" P300 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 6" P300	2
3	STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300	4
4	FLANGE WELD-NECK 6" P300 RAISED FACE	2
5	PIPE 6" P300 BEVELLED ENDS	69'-11-1/16"
6	ELBOW 90° R = 5/3 OD 6" P300	5
7	TEE REDUCING 6" × 3/4" P300	1

WELDS 15 (2 FIELD) · SUPPORTS 3 · SPOOLS 3

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15F

6"-P300-P-1012

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
E-103-N4 → D-101-N1
PIPE 69'-11-1/16" · WEIGHT 1498 lb

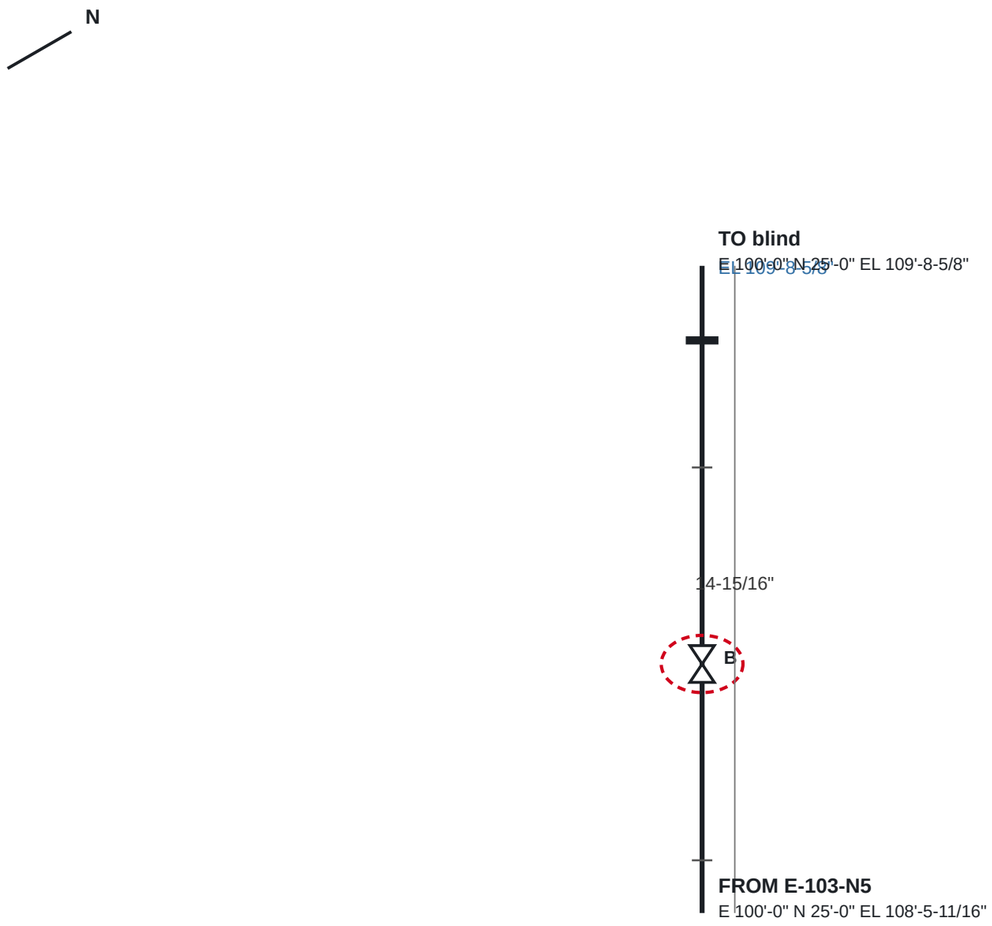
DWG ISO-6"-P300-P-1012 REV 0 WO-300-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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Isometric ISO-1"-P150-V-6027

ISOMETRIC — 1"-P150-V-6027 — Pull, clean and test the tube bundle of E-103



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-V-6027

CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
E-103-N5 → blind
PIPE 0.000" · WEIGHT 39 lb

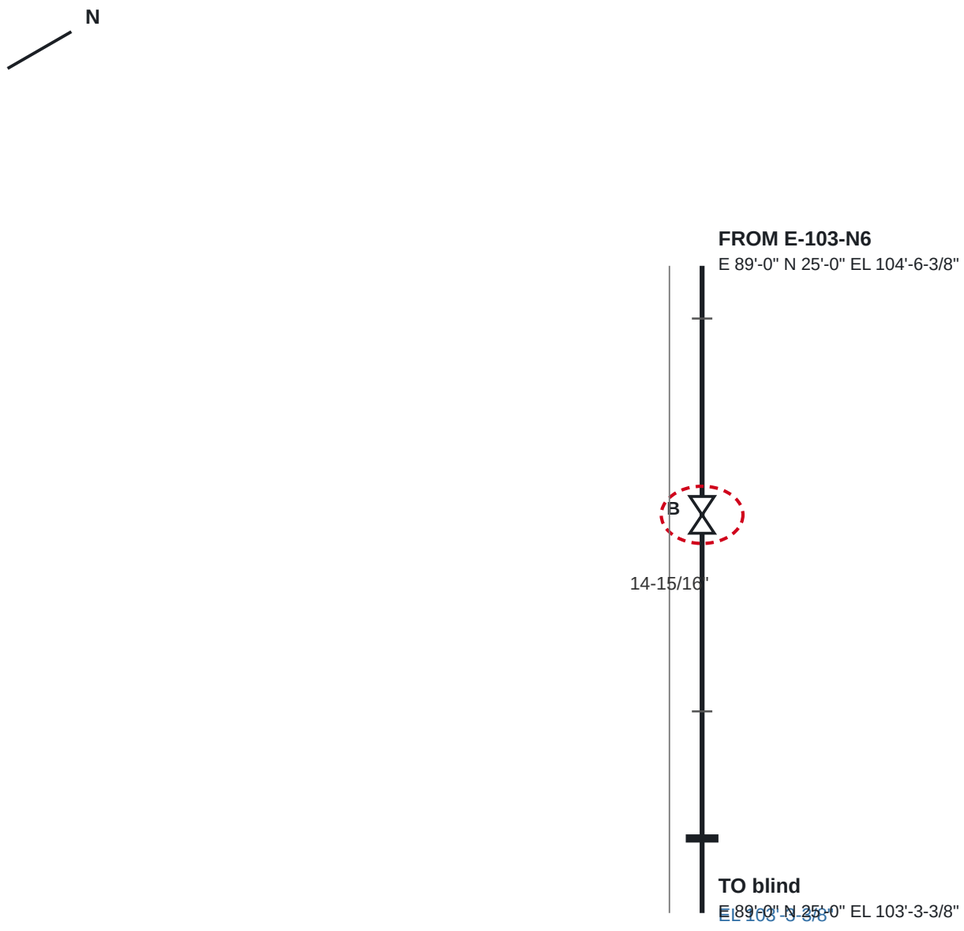
DWG ISO-1"-P150-V-6027 REV 0 WO-300-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-1"-P150-D-6028

ISOMETRIC — 1"-P150-D-6028 — Pull, clean and test the tube bundle of E-103



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-D-6028

CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
E-103-N6 → blind
PIPE 0.000" · WEIGHT 39 lb

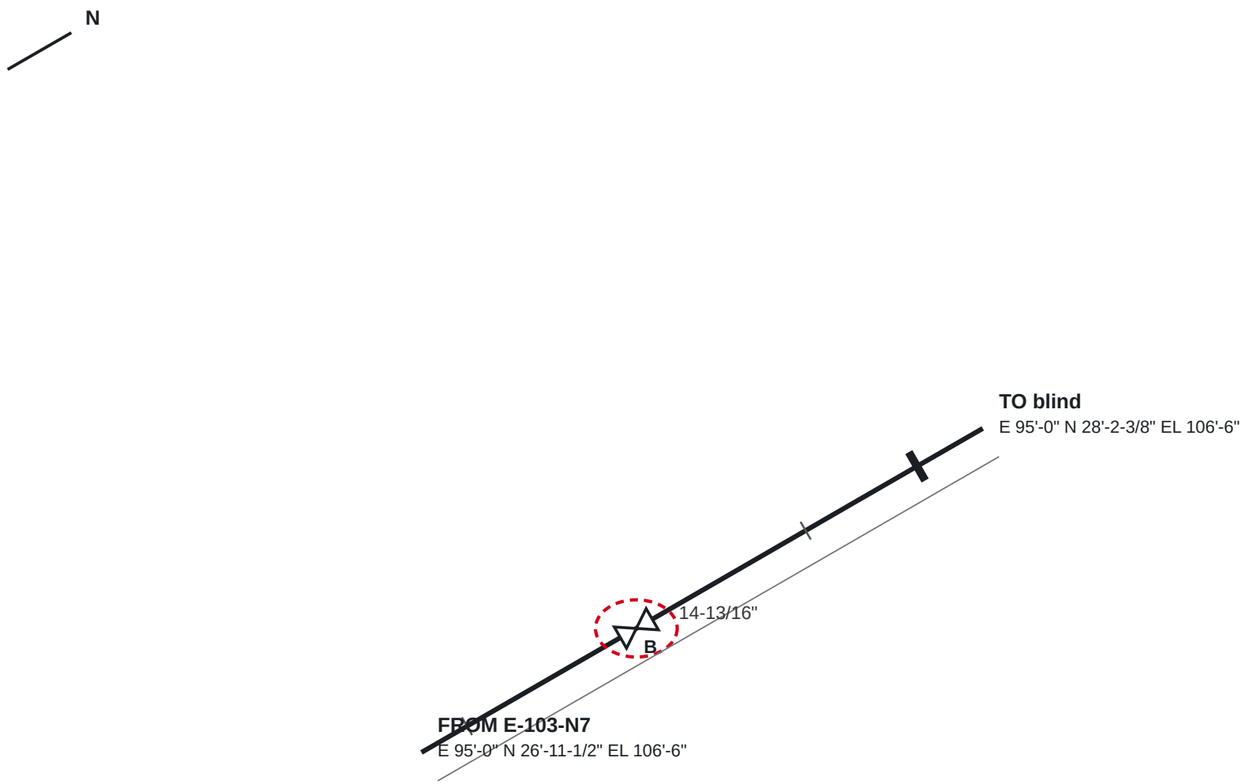
DWG ISO-1"-P150-D-6028 REV 0 WO-300-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE

Isometric ISO-3/4"-P300-I-6029

ISOMETRIC — 3/4"-P300-I-6029 — Pull, clean and test the tube bundle of E-103



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-I-6029

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
E-103-N7 → blind
PIPE 0.000" · WEIGHT 32 lb

DWG ISO-3/4"-P300-I-6029 REV 0 WO-300-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE